

CONTACT INFORMATION

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EDUCATION

The University of Texas at Austin, Austin, TX, USA. 2022-2027
 Ph.D. Candidate in Statistics at the Department of Statistics and Data Science.

- Advisors: Professor [Nhat Ho](#) and Professor [Alessandro Rinaldo](#).
- GPA: 4.0/4.0

Ho Chi Minh University of Science, Ho Chi Minh City, Vietnam. 2017-2020
 Bachelor of Science in Mathematics and Computer Science, Honor Program.

- Advisor: Professor [Dang Duc Trong](#).
- GPA: 9.75/10.0 - *Summa Cum Laude*.

RESEARCH EXPERIENCE

The University of Texas at Austin, Austin, TX, USA. 2024-2026
 Graduate Research Assistant.

- Research topic: Statistical Foundation of Mixture-of-Experts models
- Supervisors: Professor [Nhat Ho](#) and Professor [Alessandro Rinaldo](#).

Morgan Stanley, New York, NY, USA. Summer 2026
 Summer Associate.

- Research topics: Mixture of Experts in Low-rank Adaptation for Language Models.

Microsoft Corporation, Redmond, WA, USA. Summer 2024
 Research Intern.

- Research topics: Applications of Mixture of Experts in Language Models.

VinAI Research (acquired by Qualcomm AI Research), Hanoi, Vietnam. 2020-2022
 AI Research Resident.

- Research topics: Optimal Transport theory and its applications in Domain Adaptation.

RESEARCH INTERESTS

My research broadly lies at the intersection of statistical machine learning and theoretical foundation for AI, with a particular focus on four important aspects of Mixture-of-Experts (MoE) models, including Foundation (statistical understandings of gating mechanism and expert estimation), Scalability (effective sparse MoE in language models), Heterogeneity (MoE in multi-modal learning), and Efficiency (MoE in parameter-efficient fine-tuning, namely low-rank adaptation and prompt-based tuning). Previously, I also worked on computational and approximation aspects of optimal transport.

PUBLICATIONS

26. Minh Le, Bao-Ngoc Dao, **Huy Nguyen**, Quyen Tran, Anh Nguyen, Nhat Ho. [One-Prompt Strikes Back: Sparse Mixture of Experts for Prompt-based Continual Learning](#). *In International Conference on Learning Representations, 2026*.
25. Minh Le, Anh Nguyen, **Huy Nguyen**, Chau Nguyen, Anh Tran, Nhat Ho. [Revisit Visual Prompt Tuning: The Expressiveness of Prompt Experts](#). *In International Conference on Learning Representations, 2026*.
24. **Huy Nguyen**, Nhat Ho**, Alessandro Rinaldo**. [Convergence Rates for Softmax Gating Mixture of Experts](#). *IEEE Transactions on Information Theory* 72(2), 1276-1304, 2026.
23. **Huy Nguyen**, Pedram Akbarian*, Trang Pham*, Trang Nguyen*, Shujian Zhang, Nhat Ho. [Statistical Advantages of Perturbing Cosine Router in Mixture of Experts](#). *In International Conference on Learning Representations, 2025*.

22. Nghiem Tuong Diep*, **Huy Nguyen***, Chau Nguyen, Minh Le, Duy Minh Ho Nguyen, Daniel Sonntag, Mathias Niepert, Nhat Ho. [On Zero-Initialized Attention: Optimal Prompt and Gating Factor Estimation](#). *Proceedings of the International Conference on Machine Learning, 2025*.
21. Tuan Truong*, Chau Nguyen*, **Huy Nguyen***, Minh Le, Trung Le, Nhat Ho. [RepLoRA: Reparameterizing Low-rank Adaptation via the Perspective of Mixture of Experts](#). *Proceedings of the International Conference on Machine Learning, 2025*.
20. Minh Le*, Chau Nguyen*, **Huy Nguyen***, Quyen Tran, Trung Le, Nhat Ho. [Revisiting Prefix-tuning: Statistical Benefits of Reparameterization among Prompts](#). *In International Conference on Learning Representations, 2025*.
19. Fanqi Yan*, **Huy Nguyen***, Dung Le*, Pedram Akbarian, Nhat Ho**, Alessandro Rinaldo**. [On Minimax Estimation of Parameters in Softmax-Contaminated Mixture of Experts](#). *Advances in Neural Information Processing Systems, 2025*.
18. Fanqi Yan*, **Huy Nguyen***, Dung Le*, Pedram Akbarian, Nhat Ho. [Understanding Expert Structures on Minimax Parameter Estimation in Contaminated Mixture of Experts](#). *In International Conference on Artificial Intelligence and Statistics, 2025*.
17. **Huy Nguyen**, Nhat Ho**, Alessandro Rinaldo**. [Sigmoid Gating is More Sample Efficient than Softmax Gating in Mixture of Experts](#). *Advances in Neural Information Processing Systems, 2024*.
16. Xing Han, **Huy Nguyen***, Carl Harris*, Nhat Ho, Suchi Saria. [FuseMoE: Mixture-of-Experts Transformers for Fleximodal Fusion](#). *Advances in Neural Information Processing Systems, 2024*.
15. Minh Le, An Nguyen*, **Huy Nguyen***, Trang Nguyen*, Trang Pham*, Linh Van Ngo, Nhat Ho. [Mixture of Experts Meets Prompt-Based Continual Learning](#). *Advances in Neural Information Processing Systems, 2024*.
14. **Huy Nguyen**, Nhat Ho**, Alessandro Rinaldo**. [On Least Square Estimation in Softmax Gating Mixture of Experts](#). *Proceedings of the International Conference on Machine Learning, 2024*.
13. **Huy Nguyen**, Pedram Akbarian, Nhat Ho. [Is Temperature Sample Efficient for Softmax Gaussian Mixture of Experts?](#) *Proceedings of the International Conference on Machine Learning, 2024*.
12. **Huy Nguyen**, Pedram Akbarian, TrungTin Nguyen, Nhat Ho. [A General Theory for Softmax Gating Multinomial Logistic Mixture of Experts](#). *Proceedings of the International Conference on Machine Learning, 2024*.
11. **Huy Nguyen**, Pedram Akbarian, Fanqi Yan, Nhat Ho. [Statistical Perspective of Top-K Sparse Softmax Gating Mixture of Experts](#). *In International Conference on Learning Representations, 2024*.
10. **Huy Nguyen***, TrungTin Nguyen*, Khai Nguyen, Nhat Ho. [Towards Convergence Rates for Parameter Estimation in Gaussian-gated Mixture of Experts](#). *In International Conference on Artificial Intelligence and Statistics, 2024*.
9. **Huy Nguyen**, Khai Nguyen, Nhat Ho. [On Parameter Estimation in Gaussian Deviated Mixture of Experts](#). *In International Conference on Artificial Intelligence and Statistics, 2024*.
8. **Huy Nguyen**, TrungTin Nguyen, Nhat Ho. [Demystifying Softmax Gating Function in Gaussian Mixture of Experts](#). *Advances in Neural Information Processing Systems, 2023 (Spotlight, Top 3.6% out of 12343 submissions)*.
7. Dat Do*, **Huy Nguyen***, Khai Nguyen, Nhat Ho. [Minimax Optimal Rate for Parameter Estimation in Multivariate Deviated Models](#). *Advances in Neural Information Processing Systems, 2023*.
6. Dung Le*, **Huy Nguyen***, Khai Nguyen*, Trang Nguyen*, Nhat Ho. [Fast Approximation of the Generalized Sliced-Wasserstein Distance](#). *IEEE International Conference on Acoustics, Speech and Signal Processing, 2024*.
5. Khai Nguyen, Tongzheng Ren, **Huy Nguyen**, Litu Rout, Tan Nguyen, Nhat Ho. [Hierarchical Sliced Wasserstein Distance](#). *In International Conference on Learning Representations, 2023*.

4. Khang Le*, Dung Le*, **Huy Nguyen***, Dat Do, Tung Pham, Nhat Ho. [Entropic Gromov-Wasserstein between Gaussian Distributions](#). *Proceedings of the International Conference on Machine Learning*, 2022.
3. Khang Le*, **Huy Nguyen***, Khai Nguyen, Tung Pham, Nhat Ho. [On Multimarginal Partial Optimal Transport: Equivalent Forms and Computational Complexity](#). *In International Conference on Artificial Intelligence and Statistics*, 2022.
2. Khang Le*, **Huy Nguyen***, Quang Minh Nguyen, Tung Pham, Hung Bui, Nhat Ho. [On Robust Optimal Transport: Computational Complexity and Barycenter Computation](#). *Advances in Neural Information Processing Systems*, 2021.
1. Thu Nguyen, Duy H. M. Nguyen, **Huy Nguyen**, Binh T. Nguyen, Bruce A. Wade. [EPEM: Efficient Parameter Estimation for Multiple Class Monotone Missing Data](#). *Information Sciences Journal*, Volume 567, page 1-22.

PREPRINTS

13. Dung Le*, **Huy Nguyen***, Trang Pham, Alessandro Rinaldo, Nhat Ho. [Characterizing Heterogeneous Rates in Finite Mixture Estimation via Partial Optimal Transport](#). *Under review*, [arXiv:2609.16622](#).
12. **Huy Nguyen***, Dung Le*, Alessandro Rinaldo, Nhat Ho. [On the Geometry of Separation in Finite Gaussian Mixtures](#). *Under review*, [arXiv:2606.16179](#).
11. Nicola Barilleto*, **Huy Nguyen***, Nhat Ho**, Alessandro Rinaldo**. [On Bayesian Softmax-Gated Mixture-of-Experts Models](#). *Under review*, [arXiv:2604.20551](#).
10. Viet Nguyen*, Tuan Minh Pham*, Thinh Cao*, Tan Dinh, **Huy Nguyen**, Nhat Ho**, Alessandro Rinaldo**. [A Statistical Theory of Gated Attention through the Lens of Hierarchical Mixture of Experts](#). *Under review*, [arXiv:2602.01468](#).
9. Tuan Minh Pham*, Thinh Cao*, Viet Nguyen*, **Huy Nguyen**, Nhat Ho**, Alessandro Rinaldo**. [Rethinking Multinomial Logistic Mixture of Experts with Sigmoid Gating Function](#). *Under review*, [arXiv:2602.01466](#).
8. Fanqi Yan*, Dung Le*, Trang Pham, **Huy Nguyen**, Nhat Ho. [Improving Minimax Estimation Rates for Contaminated Mixture of Multinomial Logistic Experts via Expert Heterogeneity](#). *Under review*, [arXiv:2602.00939](#).
7. **Huy Nguyen**, Thong T. Doan, Quang Pham, Nghi Bui, Nhat Ho**, Alessandro Rinaldo**. [On DeepSeekMoE: Statistical Benefits of Shared Experts and Normalized Sigmoid Gating](#). *Under review*, [arXiv:2505.10860](#).
6. Nghiem T. Diep, Dung Le, Tuan Truong, Tan Dinh, **Huy Nguyen**, Nhat Ho. [Hypernetwork-Driven Low-Rank Adaptation Across Attention Heads](#). *Under review*, [arXiv:2510.04295](#).
5. Nghiem T. Diep, Hien Dang, Tuan Truong, Tan Dinh, **Huy Nguyen**, Nhat Ho. [DoRAN: Stabilizing Weight-Decomposed Low-Rank Adaptation via Noise Injection and Auxiliary Networks](#). *Under review*, [arXiv:2510.04331](#).
4. **Huy Nguyen***, Fanqi Yan*, Pedram Akbarian, Nhat Ho**, Alessandro Rinaldo**. [Sigmoid Self-Attention has Lower Sample Complexity than Softmax Self-Attention: A Mixture-of-Experts Perspective](#). *Under review*, [arXiv:2502.00281](#).
3. **Huy Nguyen***, Xing Han*, Carl Harris, Suchi Saria**, Nhat Ho**. [On Expert Estimation in Hierarchical Mixture of Experts: Beyond Softmax Gating Functions](#). *Under review*, [arXiv:2410.02935](#).
2. Pedram Akbarian*, **Huy Nguyen***, Xing Han*, Nhat Ho. [Quadratic Gating Mixture of Experts: Statistical Insights into Self-Attention](#). *Under review*, [arXiv:2410.11222](#).
1. Nam V. Nguyen, **Huy Nguyen**, Quang Pham, Van Nguyen, Savitha Ramasamy, Nhat Ho. [CompeteSMoE - Statistically Guaranteed Mixture of Experts Training via Competition](#). *Under review*, [arXiv:2505.13380](#).

TALKS

7. Mixture of Experts in Large-scale and Multimodal Models. *Morgan Stanley Research Talk, New York, 2026 (Invited talk)*.
6. FuseMoE: Mixture-of-Experts Transformers for Fleximodal Fusion. *Jane Street Graduate Research Fellowship Workshop, New York, 2026 (Invited talk)*.
5. Mixture of Experts in Large-scale and Multimodal Models. *MIT reading group hosted by Prof. Priya Donti, Virtual, 2025 (Invited talk)*.
4. Mixture of Experts in Large-scale and Multimodal Models. *Two Sigma PhD Fellowship Finalist Reception, Virtual, 2025 (Invited talk)*.
3. Mixture of Experts in Large-scale and Multimodal Models. *Dartmouth Applied and Computational Mathematics Seminar, Virtual, 2025 (Invited talk)*.
2. Demystifying Softmax Gating Function in Gaussian Mixture of Experts. *STATML@UT Reading Group, Austin, TX, 2024 (Invited talk)*.
1. Demystifying Softmax Gating Function in Gaussian Mixture of Experts. *IFML Workshop on Generative AI (Student talks session), Austin, TX, 2023 (Invited talk)*.

TEACHING EXPERIENCE

The University of Texas at Austin, Austin, TX, USA.

Teaching Assistant at the Department of Statistics and Data Sciences.

- SDS302F - Foundations of Data Analysis. Fall 2022
- SDS322E - Elements of Data Science. Spring 2023
- SDS320E - Elements of Statistics. Fall 2023
- SDS315 - Statistical Thinking. Spring 2025

PROFESSIONAL SERVICES

Senior Program Committee/Area Chair

- the Association for the Advancement of Artificial Intelligence ([AAAI](#)) 2027.

Program Committee/Reviewer at

- the Electronic Journal of Statistics ([EJS](#)).
- the Journal of Machine Learning Research ([JMLR](#)).
- the IEEE Transactions on Pattern Analysis and Machine Intelligence ([TPAMI](#)).
- the Transactions on Machine Learning Research ([TMLR](#)).
- the International Conference on Machine Learning ([ICML](#)) 2022-2026.
- the Conference on Neural Information Processing Systems ([NeurIPS](#)) 2022-2026.
- the International Conference on Artificial Intelligence and Statistics ([AISTATS](#)) 2022-2026.
- the International Conference on Learning Representations ([ICLR](#)) 2024-2027.
- the Association for the Advancement of Artificial Intelligence ([AAAI](#)) 2025-2026.

Co-organizer of the Statistical Machine Learning seminar at UT Austin ([STATML@UT](#)).

PROFESSIONAL MEMBERSHIP

- Regular Member of the Institute of Electrical and Electronics Engineers (IEEE).
- Student Member of the Institute of Mathematical Statistics (IMS).

HONORS AND AWARDS	• UT Austin Outstanding Graduate Research Fellowship.	2026
	• UT Austin Professional Development Award.	2024, 2025
	• ICML 2026 Silver Reviewer.	2026
	• Two Sigma PhD Fellowship Finalist.	2025
	• Top Reviewer at NeurIPS 2024.	2024
	• AISTATS 2024 Registration Grant.	2024
	• ICLR 2024 Travel Award.	2024
	• NeurIPS 2023 Scholar Award.	2023
MENTORSHIP	• Viet Nguyen , <i>Undergrad student, Hanoi University of Science and Technology</i>	
	• Minh Pham, <i>Undergrad student, Purdue University</i>	
	• Thinh Cao , <i>Undergrad student, Hanoi University of Science and Technology</i>	
	• Tan Dinh , <i>Undergrad student, University of Massachusetts at Amherst</i>	
REFERENCES	• Nhat Ho . Email: minhnhhat@utexas.edu (Advisor).	
	• Alessandro Rinaldo . Email: alessandro.rinaldo@austin.utexas.edu (Advisor).	